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ISSS602 Data Analytics Lab

Assignment 2: Hospitality Segmentation with Cluster Analysis

Segment Pricing & Risk on the Northern Adriatic Coast

Supply-side clustering of coastal accommodations in Veneto & Emilia-Romagna

An executive summary done by Nor Hendra



Clustering Metrics, Univariate and Bivariate Analysis

Geospatial Proximity

Distance to Sea (z-scored dist_sea): Primary coastal geometry signal that captures both open-coast amenity appeal and surge-exposure pathway in one compact measure.

Distance to Waterways (z-scored dist_waterways): Distinct hydrologic setting (canals/lagoon/river backflow) not redundant with open-sea proximity; adds new spatial signal.

Elevation (z-scored grond_z): Vertical position that, together with sea-level dynamics, shapes vulnerability gradients; include as a core geospatial driver.

Physical Infrastructure

Building Area (log transformed and z-scored area): Proxy for asset scale and on-site capacity; log transform tames heavy right tail so very large complexes don't dominate distances.

Building Height (z-scored height): Structural verticality relevant to perceived quality and operability during shallow inundation.

Business Attributes

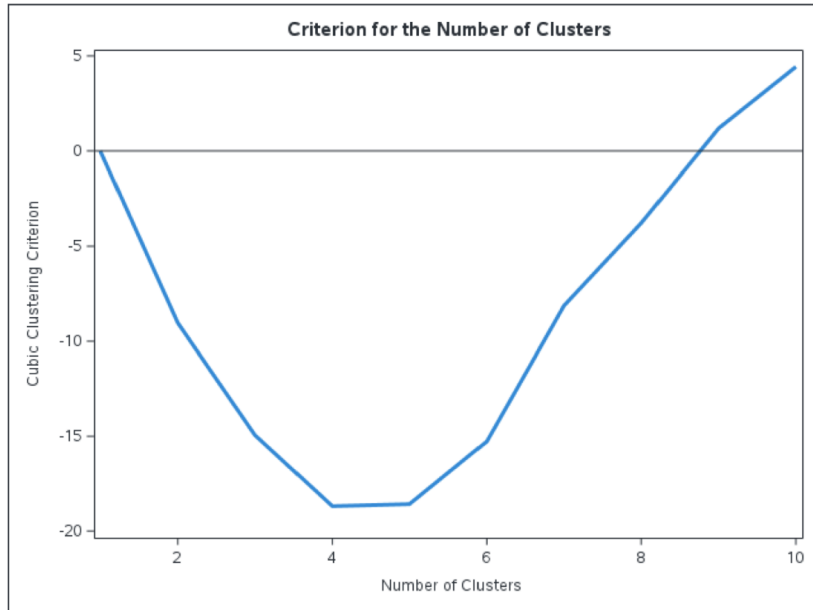
Price (log transformed and z-scored price): Realised market position/pricing power; log stabilises skew while preserving rank so premium vs value supply separates cleanly.

Environmental Features (Profiling only)

Beach access, room type, and pool access are excluded from the clustering distance and reserved for post-hoc profiling/validity checks to avoid redundancy and because they are nominal (better handled outside Euclidean space).



Hierarchical Clustering



Cubic Clustering Criterion

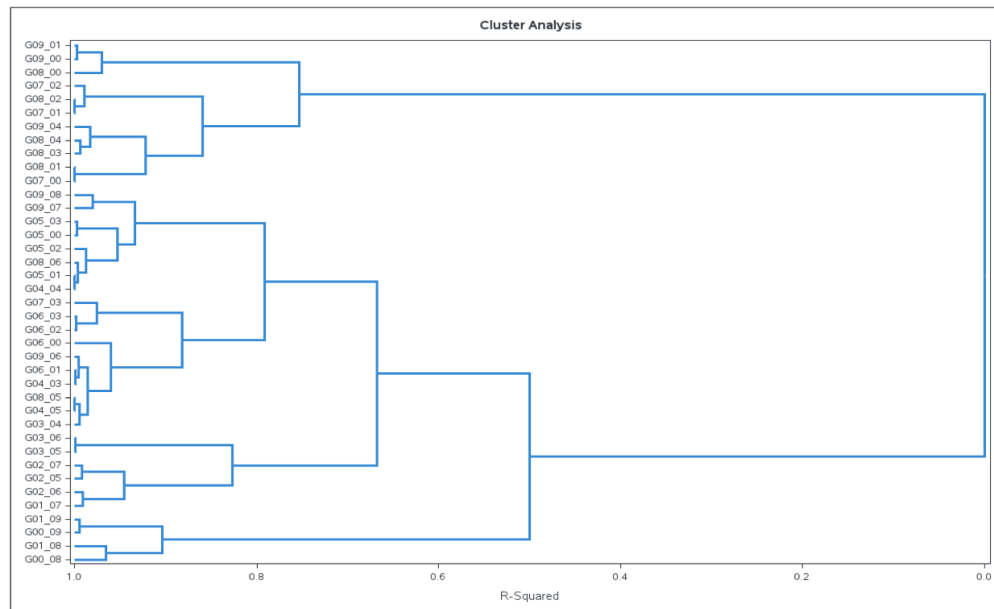
The CCC bottoms around $k \approx 4 - 5$ and then rises, turning positive only at high $k (\approx 9 - 10)$.

Dendrogram

To make the tree interpretable, properties were first binned into a 10x10 latitude-longitude grid and each dendrogram leaf labeled by its grid ID. Leaves were annotated using each region's median z-score for distance-to sea and distance-to-waterways, assigning one of three observed environmental regimes: (1) Near Sea & Waterways, (2) Near Sea & Away from Waterways, (3) Away from Sea & Near Waterways. The grouped labeled- reports 17 regions with 43.6%, 25.6%, and 30.8% in the regimes. Ultimately indicating the study area is dominated by coastal-waterway adjacency with substantial open-coast and inland-waterway representation.

Selecting $k = 5$

The regime-level separation (coastal-waterway adjacency, open-coast, inland-waterway) is preserved, which matters for amenity appeal vs. exposure positioning. Additional splits within regimes (driven by area/height/price) support distinct offerings. This will also allow to retain those contrasts while allowing within-regime differentiation by size/price.

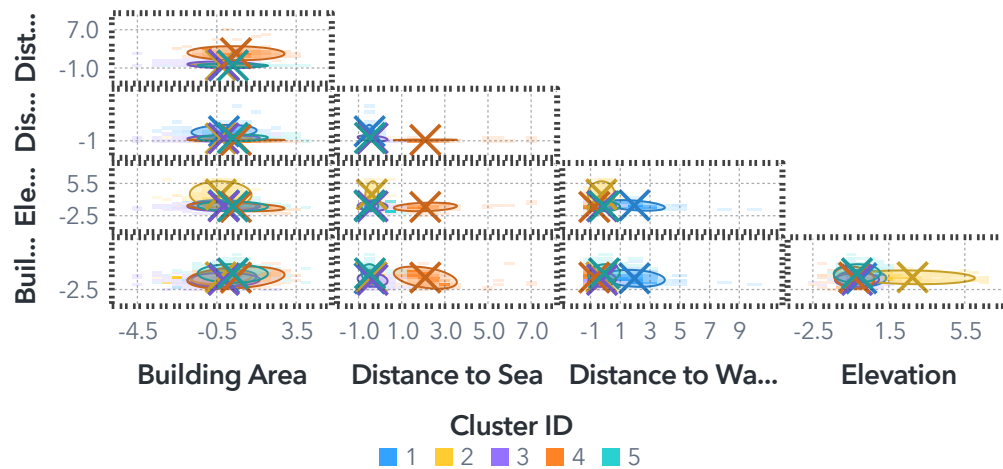


No.	Proximity Tag	Region IDs
1	Away from Sea & Near Waterways	G07_00, G07_01, G07_02, G07_03, G08_00, G08_01, G08_02, G08_03, G08_04, G09_00, G09_01, G09_04
2	Near Sea & Away from Waterways	G01_07, G02_05, G02_06, G02_07, G03_05, G03_06, G05_00, G05_03, G09_07, G09_08
3	Near Sea & Near Waterways	G00_08, G00_09, G01_08, G01_09, G03_04, G04_03, G04_04, G04_05, G05_01, G05_02, G06_00, G06_01, G06_02, G06_03, G08_05, G08_06, G09_06

K-Means Clustering

Cluster

Observations: **3K of 3K** Polylines: **689**



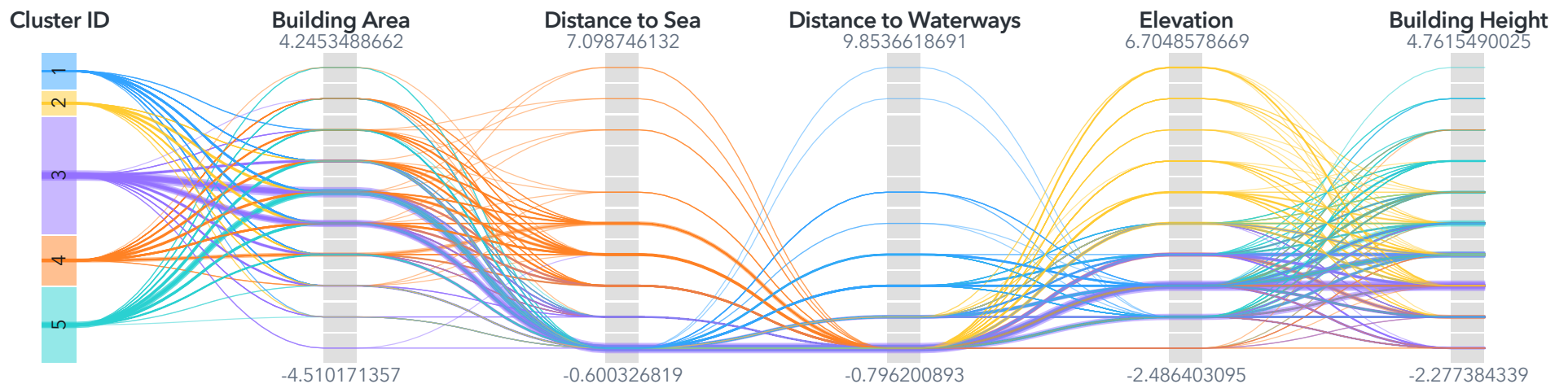
In the cluster diagram, clusters separate most clearly along Distance to Sea and Elevation, with Distance to Waterways providing a secondary axis of differentiation; Building Area and Building Height act mainly as within-regime refiners (their ellipses overlap more), which is exactly what we want given the variable design. The parallel-coordinates plot reinforces this: each cluster traces a distinct multi-dimensional pattern, with some segments simultaneously near the sea and low-to-moderate elevation, others farther from the sea but near waterways and on higher ground, and others distinguished by larger area / taller structures.

The five-segment solution preserves the environmental regimes revealed earlier by Ward's hierarchy (coastal-waterway interface, open-coast, inland-waterway) while adding separation by asset scale and pricing.

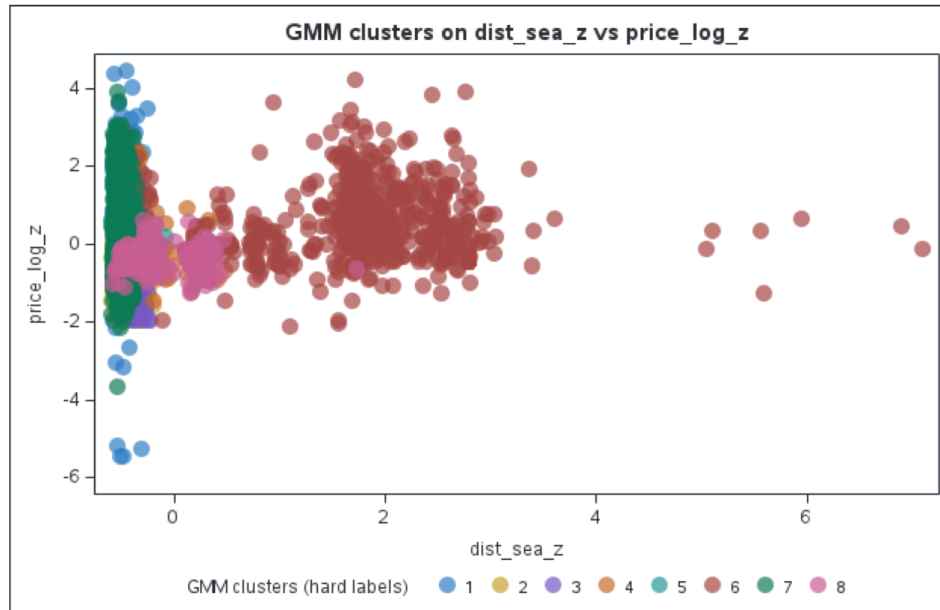
- Cluster 1** near sea but away from rivers, value low-rise.
- Cluster 2** high-ground coastal, small/value.
- Cluster 3** waterfront budget low-rise.
- Cluster 4** canal-front inland premium with larger footprints.
- Cluster 5** beach-near premium high-rise.

Cluster

Observations: **3K of 3K** Polylines: **689**



Gaussian Mixture Model



Cluster sizes are uneven, most observations fall in C7 (36.5%) and C6 (19.7%), while C5 is tiny (0.27%).

Price vs. sea-distance: “near-sea” is not a guarantee of premium. Among clusters with negative dist_sea_z (nearer than average to the sea), price signals split both ways:

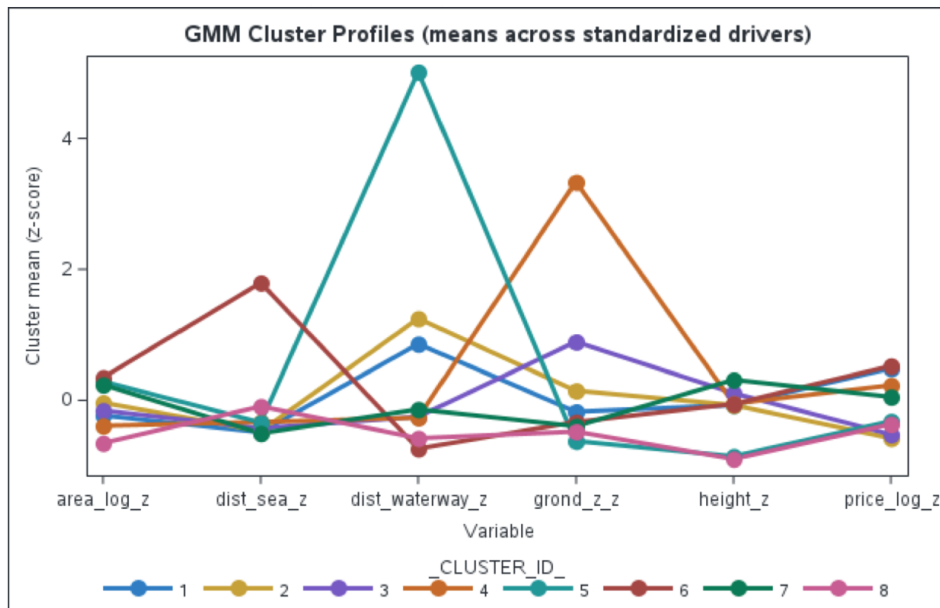
Premium near-sea: C1 (price +0.487 SD) and C4 (+0.229 SD).

Neutral near-sea: C7 (price +0.053 SD).

Discount near-sea: C2 (−0.579 SD), C3 (−0.522 SD), C8 (−0.364 SD).

Premium can also be inland, conditional on other features. C6 sits farther from the sea on average (dist_sea_z ≈ +1.79) yet shows a positive price (+0.525 SD) with larger area and near-waterway siting (negative dist_waterway_z).

Posterior max-probability is high: median = 0.9158, IQR 0.7569–0.9844 (25th–75th), max= 1.0000 which indicates **confident assignments overall**. The radar chart (cluster means across standardized drivers) shows that geography organizes the market and form-factor stratifies price within it: clusters split first by sea and waterway proximity and by elevation, then diverge on area/height and price. Overall, the radar confirms that sea proximity alone is insufficient; near-sea + size/height (sometimes aided by waterway adjacency) is what consistently lifts price, while near-sea without those attributes settles into value tiers.





Cluster 1 (12.3%, N=364) - Near-sea, away from waterways, budget low-rise.

This segment is closest to the sea ($\text{dist_sea_mean} \approx -0.48$) but farthest from major canals ($\text{dist_waterway} \approx +1.96$), indicating properties along open coastlines without nearby waterways. It has a **value-oriented profile**, price is about 0.45 standard deviations below the mean, and buildings are relatively low-rise ($\text{height} \approx -0.14$). Area is slightly below average (-0.14) and elevation is around the mean ($+0.05$). This cluster likely represents smaller, inexpensive accommodations on the beach fringe without direct canal access. The cluster is fairly compact (average distance to centroid ~ 1.61 , with a few outliers reaching ~ 8.1) indicating homogeneous characteristics among members.

Cluster 2 (8.3%, N=246) - High-ground coastal, small, value.

The defining feature is extremely high elevation ($\text{grond_z} \approx +2.75$), meaning these properties sit on much higher ground than others despite being near the coast. They are **close to the sea** ($\text{dist_sea} \approx -0.39$) and moderately close to waterways ($\text{dist_waterway} \approx -0.28$) - perhaps hilltop or high dune locations not far from water. These tend to be **small and lower-priced**: $\text{area} \approx -0.31$, $\text{price} \approx -0.38$, $\text{height} \sim +0.09$ (essentially average height). We might interpret Cluster 2 as a niche of small, budget-friendly accommodations benefiting from elevated (safer) coastal positions. Consistent with a niche, it was one of the looser clusters in k-means (avg distance ~ 1.85), suggesting a bit more internal variation.



Cluster 3 and Cluster 4



Cluster 3 (38.0%, N=1,125) – Waterfront budget low-rise.

This is the largest cluster, encompassing over a third of the listings. These properties are **close to both the sea and waterways** ($\text{dist_sea} \approx -0.35$, $\text{dist_waterway} \approx -0.22$), indicating they are in the heart of the lagoon or waterfront zones. They have the **lowest price** on average ($\text{price} \approx -0.50$, a significant discount) and the shortest buildings ($\text{height} \approx -0.45$, decidedly low-rise). Area is slightly below average (-0.16) and elevation is a bit below average (-0.17), which could reflect low-lying waterfront locales. This cluster represents the **budget accommodations in prime waterfront locations** – likely older or simpler properties that do not command high prices despite their proximity to water. It was the most compact cluster (avg intra-cluster distance ~ 1.38), indicating very consistent attributes, and appropriately so given its clear profile.

Cluster 4 (16.5%, N=488) – Canal-front inland premium, larger footprint.

This group is characterized by being the **farthest from the sea** of all segments ($\text{dist_sea} \approx +2.10$) while being **very close to waterways** ($\text{dist_waterway} \approx -0.74$). In other words, these listings are situated inland along canals/rivers. They tend to have a **larger area** ($\text{area} \approx +0.47$) and command an **above-average price** ($\text{price} \approx +0.64$), suggesting more spacious properties that leverage their canal-front charm for a premium price. Building height is around average ($+0.01$) and elevation is slightly below average (-0.33), implying many are on normal ground (not elevated) and mid-rise constructions. Despite the diverse geography, the cluster's definition (inland waterway premium properties) holds together, though it's among the looser clusters (avg distance ~ 1.87). Cluster 4 stands apart as the only cluster lacking direct sea proximity yet maintaining high prices through canal-side appeal and size.





Cluster 5 (24.9%, N=735) – Beach-near premium high-rise.

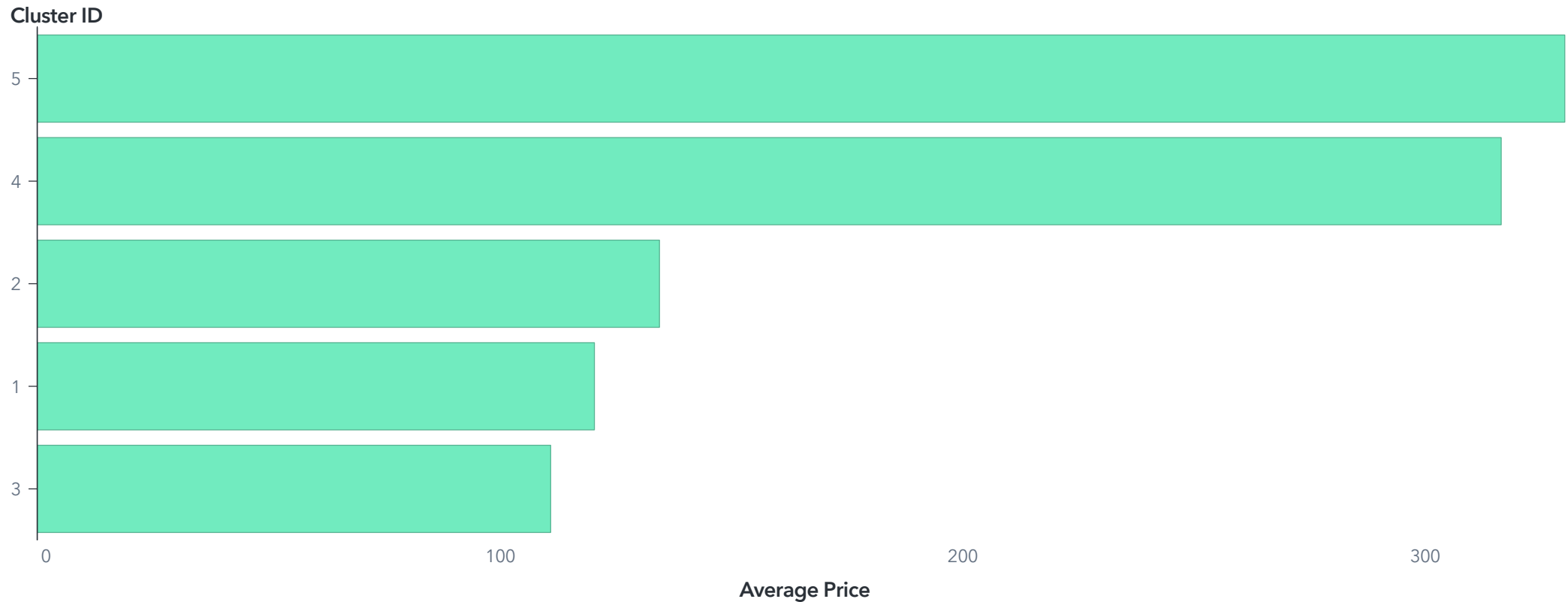
This segment features **coastal properties with the tallest buildings**. Distance to sea is negative ($\text{dist_sea} \approx -0.47$, i.e. near the sea) and distance to waterways is modestly below average ($\text{dist_waterway} \approx -0.11$), so many have both beach proximity and some canal presence, though not as distinctly canal-front as Cluster 4. The standout driver is building height: $\text{height} \approx +0.90$, indicating these are high-rise or multi-story structures. They also have **above-average area** ($\text{area} \approx +0.31$) and **premium pricing** ($\text{price} \approx +0.54$). Elevation is slightly below average (-0.18), consistent with being near sea level. Cluster 5 can be viewed as the **upmarket coastal towers** – likely hotels or larger resorts near the beachfront. It is fairly compact (avg intra-distance ~ 1.67). Interestingly, its nearest neighbor cluster in the feature space is Cluster 3 (centroid distance ~ 1.78), indicating that the only major difference between Cluster 5 and Cluster 3 is the *quality/form-factor* (premium high-rise vs. budget low-rise) while both share the near-water geography. This underscores a **continuum from budget to premium within the near-sea market**, rather than completely disjoint segments.

Summary

Overall, the five k-means clusters cover the full coastal accommodation spectrum identified in the hierarchical step. The cluster sizes range from about 8% to 38% of the sample, indicating a relatively balanced segmentation without any segment dominating unrealistically. Quality metrics confirm the solution's validity: within-cluster **compactness** (average distance from members to centroid) ranges from 1.38 to 1.87, and the **separation** between cluster centroids ranges from ~ 1.78 (closest pair) to ~ 2.98 (furthest pair). The tightest cluster is C3 (waterfront budget) and the loosest is C4 (inland premium), which is expected given C4's more heterogeneous inland coverage. The nearest centroids being C3 vs C5 reflects the earlier point that budget and premium near-water groups form a gradual continuum. Meanwhile, the largest centroid distance (C2 vs C3) highlights how distinct the high-ground coastal niche is from the low-lying waterfront budget segment. In sum, the k-means analysis at $k = 5$ not only corroborates the cluster structure hinted by Ward's method, but also provides clear **profiles for each segment** in terms of geography and price drivers. Each cluster corresponds to a meaningful market segment – for example, an inland canal-side premium segment versus a beachfront budget segment – which can be mapped back to real-world locations and property characteristics. This gives confidence that the chosen segmentation is both data-driven and interpretable for stakeholders.

Price Segment (Average Price of Cluster IDs)

Average Price by Cluster ID



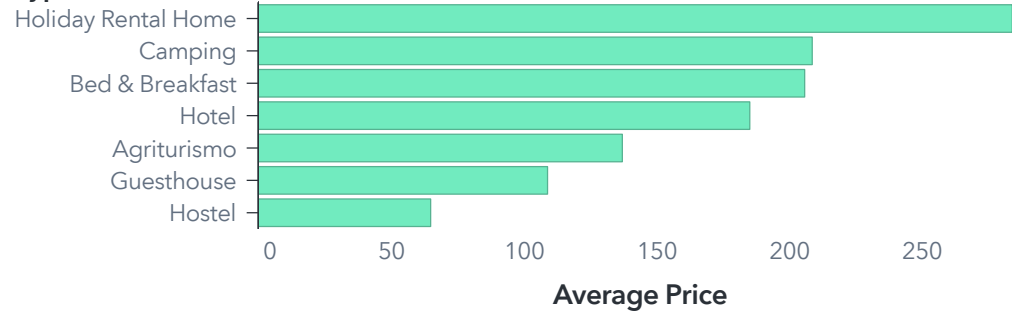
The bar chart is consistent with the clustering and working labels, showing compositional levers behind the monetisation. Cluster 5 ("beach-near premium high-rise") and Cluster 4 ("canal-front inland premium, larger footprint") monetise at the top end, while Clusters 1/3 ("budget low-rise" near sea/waterfront) anchor the lower tiers, and Cluster 2 ("high-ground coastal, small, value") sits in between.

This pattern is consistent with your GMM evidence that **sea proximity alone does not guarantee a premium**: the "near-sea cloud" contains both premium and discount clusters; premiums emerge when **near-sea is paired with larger area/greater height (and sometimes waterway adjacency)**.

Product Type & Access Features

Average Price by Type of Room

Type of Room

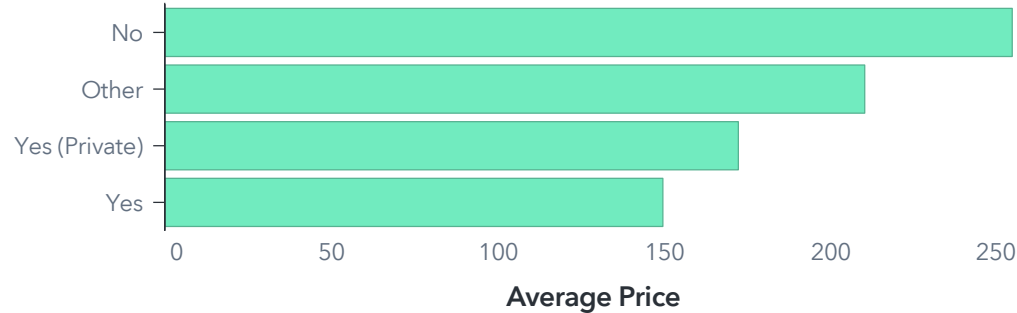


Type of Room:

Mean price is highest for Holiday Rental Homes (≈ 284), then Camping (≈ 209) and B&B (≈ 206), with Hotels lower (≈ 185). Hostels sit at the bottom (≈ 65). Sample sizes are large for Hotels ($n=2,269$) and Holiday Rental Homes ($n=547$) but very small for Agriturismo ($n=4$) and Hostel ($n=6$), so treat those two with caution. A one-way ANOVA across Types is significant ($F \approx 9.91$, $p \approx 7.44 \times 10^{-11}$), indicating real differences in mean price between types.

Average Price by Beach Access

Beach Access

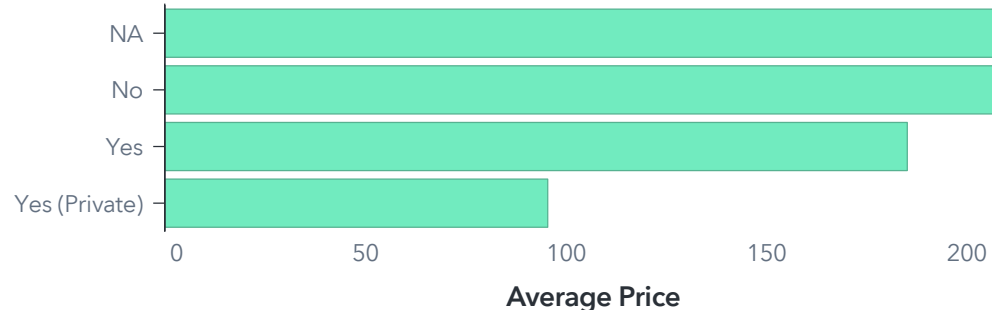


Beach Access:

The mean is highest under "No" beach access (≈ 255), then "Other"/NA (≈ 210), "Yes (Private)" (≈ 172), and "Yes" (≈ 150). This tells us that direct beach access isn't the only, or even primary price driver in this dataset; inland premium or canal-front locations (matching Cluster 4 label) likely command high prices without formal "beach access." Differences are statistically significant ($F \approx 42.66$, $p \approx 5.46 \times 10^{-19}$).

Average Price by Pool Access

Pool Access



Pool Access:

No pool and NA categories (≈ 209 - 212) average higher than shared pool (≈ 185), while private pool averages lowest (≈ 96) but with very small $n=19$, so that estimate is unstable. Differences are statistically significant but modest ($F \approx 3.22$, $p \approx 0.040$). Given the tiny private pool sample, it's prudent not to over-interpret that category without more data.

Conclusion

From a destination-management standpoint, the patterns dovetail with the region's physical risk setting. Relative sea level rise in the northern Adriatic is projected to be substantial this century; accounting for local subsidence, studies indicate ~0.81 m (likely 0.58–1.10 m) under RCP 8.5 by 2100 for the northern Adriatic (EGU, 2021), which encompasses Venice and the Romagna coast. Subsidence makes a large, location-dependent contribution: historical analyses attribute ~57% of observed RSLR at Venice and ~85% at Ravenna to land subsidence in the 20th century, compounding eustatic rise. (NHESS, 2021) (Carbognin, Teatini, & Tosi, 2009) Storm-surge dynamics (sirocco-driven setups, seiches) further amplify extremes and show trends toward higher surge components in the most severe events. These external facts corroborate why *exposure geometry* (sea/canal proximity, elevation) is the first splitter in the clusters and why business continuity hinges on micro-location and built form.

Taken together, the supply of accommodations along the Veneto and Emilia-Romagna coast is not a single coastal premium continuum. Rather, it partitions into (i) premium high-rise/coastal-towers near the sea; (ii) premium, larger-footprint canal-front inland stock; (iii) two substantial near-water value segments; and (iv) a smaller high-ground coastal value niche. This supports using the 5-segment view as the backbone for managerial planning because it explains both price realisation and exposure geometry, two pillars of destination viability in a climate-stressed coast.

References

EGU. (2021). *European Geosciences Union*. Retrieved from Sea-level rise in Venice: historic and future trends: <https://nhess.copernicus.org/articles/21/2643/2021/nhess-21-2643-2021.html>

NHESS. (2021). *Extreme floods of Venice: characteristics, dynamics, past and future evolution (review article)*. Retrieved from Natural Hazards and Earth System Sciences: <https://nhess.copernicus.org/articles/21/2705/2021/>

Carbognin, L., Teatini, P., & Tosi, L. (2009). *WIT Transactions on Ecology and the Environment*. Retrieved from <https://www.witpress.com/Secure/elibrary/papers/RAV09/RAV09012FU1.pdf>

Actionable Recommendations

Wave and storm exposure: Hotels on unsheltered reaches face higher runup and overtopping. Add long-term 99th-percentile significant wave height and/or extreme wave climate. Upon research, this might be readily available from the Copernicus Marine Mediterranean wave products (Copernicus Marine Service, 2021).

Land subsidence: Relative SLR at the property is elevation change minus sea-level change. Northern Adriatic coasts (Veneto/Po Delta) are well known for subsidence; add vertical land motion from the European Ground Motion Service (EGMS) to quantify mm/yr sinking at or near each asset, then roll it into relative SLR rates. This has become straightforward since EGMS went operational. (Copernicus, n.d.)

Shoreline change & beach width: Distance to water misses whether the shoreline is *advancing or retreating*. Add a shoreline change rate (m/yr) and beach width (peak season). Something like EU datasets derived from EUROSION (European Environment Agency, 2024) and the newer CoasTER database that integrates erosion indicators on shoreline baseline; they're European-scale and defensibly curated. Even a coarse sign (erosion vs accretion) improves separability.

Accessibility (time, not distance): Replace or complement straight-line distances with travel time to the nearest beach access point, transit stop, town center, and nearest intercity rail station, computed on the road/walk network. These time-based variables are much more predictive for pricing and clustering than Euclidean distances.

Replace the duplicative “estuary distance” with a geomorphic class: `dist_estuary` $\approx$ `dist_sea` for three-quarters of rows, switch to a coastal-setting categorical (open beach / lagoon or canalized back-beach / river mouth / hard-defended). It's a single, interpretable splitter that captures different hazard pathways (overtopping vs backflow) and avoids double-counting distance.

Compact, cluster-friendly: Clustering that reflects destination attractiveness without leaving the supply-side frame benefits from simple, reproducible proxies. Replace some straight-line distances with network travel times (e.g., to the nearest beach access point and intercity rail); add a small set of amenity densities (counts of restaurants, pharmacies, ATMs, marinas within 500–1,000 m from OSM). These measures are compact, auditable, and improve separation between “high-rise premium” and “waterfront budget” segments that happen to lie equally close to the sea.